

Susanna Bardini is a PhD student in Information Technology at Politecnico di Milano. Her research focuses on adaptive systems that integrate smart sensing technologies, heterogeneous IoT networks, and microcontroller-based platforms for personalized health and smart homes. She is particularly interested in full-stack infrastructures that combine physiological signals from wearables with ambient sensor data, leveraging real-time data fusion and federated learning on 5G-enabled edge computing frameworks

LANGUAGES

Italian: native language
English: fluent (Level C1)

COMPUTER SKILLS

Prog lang: C, C++, Python, MATLAB, R, Swift, SQL

OS usage: MacOS, Windows, Ubuntu, Raspbian

Various: Git, L^AT_EX, MySQL

Tool: Arduino IDE, EAGLE, SOLIDWORKS, FreeFem, OpenSim, MPLAB X IDE, PSoC Creator

COMMUNICATION SKILLS

Conference: EMBC (2026), Star (2024-25), EDUCON (2025), ECSS (2025)

Presentation: ETH (2026), KAUST (2026), ETH (2026), MIT Sloan (2025), BU (2025), Northwestern (2025), UIC (2025), DPI (2025)

STARTUP COMPETITION

TEF – Ignition (2026)
Switch2Product - Top 25 (2024)

MEMBERSHIPS

IEEE Student Member,
Women in Engineering Vice-Chair

EDUCATION

2024 – Present **Ph.D. Student in Information Technology**, *Politecnico di Milano*

Research focus: smart sensing technologies

Ph.D. Visiting Student, *ETH Zurich* (Mar. – Aug. 2026)

2021 – 2023 **M.Sc. Biomedical Engineering**, *Politecnico di Milano*

Thesis: “OptoSens: an Optical Sensor Based Mask for Sleep Quality Monitoring”

Fields: electronics design for biomedical instrumentation, technologies for sensors and clinical instrumentation, neuroengineering

2018 – 2021 **B.Sc. Biomedical Engineering**, *Politecnico di Milano*

Thesis: “BIONET: Biomedical Image Semantic Segmentation Via Deep Neural Network”

EXPERIENCE

Teaching Assistant, *Politecnico di Milano* (Sept. 2024 – Present)

Fondamenti di Informatica, course by Prof. Bolchini

Instructor, *Politecnico di Milano* (Sept. 2024 – Present)

Passion in Action: Arduino Fundamentals, Machine Learning for Time Series Analysis, Federated Learning Fundamentals, course by Prof. Santambrogio and Prof. Torrebruno

Teaching Assistant, *Politecnico di Milano* (Feb. 2024 – July 2024)

Capstone Projects, course by Prof. Conci

Extracurricular Intern, *Politecnico di Milano* (Jan. 2024 – Nov. 2024)

I worked in the NECSTLab of Politecnico di Milano proceeding on the work done during my master thesis and tutoring bachelor students

SELECTED PUBLICATIONS

Isella, A., **Bardini, S.**, & Santambrogio, M. D. WARM: Warm-up Readiness from Multivariate Wearable Physiology. Accepted, to appear in 48th Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC) 2026.

Ginestretti, L., **Bardini S.**, Guarnerio, A., Varone, S., & Santambrogio, M. D. Embedding-Based Representation Learning for Personalized Stress Assessment from Wearable Time Series. Accepted, to appear in 48th Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC) 2026.

Bardini S., Verosimile, A., Vaccarino G., & Santambrogio, M. D. IGEA: a Framework for Sleep Tracking via Edge-based Federated Learning on Multimodal Data. Accepted, to appear in 8th Workshop on Parallel AI and Systems for the Edge (PAISE - IPDPS) 2026.

Ginestretti, L., Lazzari, J., Verosimile, A., **Bardini, S.**, & Santambrogio, M. D. (2025, October). Towards a Machine Learning Model for Cognitive Performance Prediction. In 2025 IEEE EMBS International Conference on Biomedical and Health Informatics (BHI) (pp. 1-7). IEEE.

Bardini, S., Biondi, F., Campo B., Di Sarno, B., Crippa, L. V., & Santambrogio, M. D. Optimizing Athletic Performance Through Personalized Sleep Monitoring. Accepted, to appear in IEEE International Workshop on Sport, Technology and Research (STAR) 2025.

Berzoini, R., Colombo, A. A., **Bardini, S.**, Conelli, A., D'Arnese, E., & Santambrogio, M. D. (2022, July). An Optimized U-Net for Unbalanced Multi-Organ Segmentation. In 2022 44th Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC) (pp. 3764-3767).